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ARTICLE

A Large-Eddy Simulation Model for Boundary-Layer Flow Over Surfaces with Horizontally Resolved but Vertically Unresolved Roughness Elements

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Abstract Accurate prescription of ground-level boundary conditions is a key requirement for large-eddy simulation (LES) of atmospheric boundary-layer (ABL) flow. When the lower boundary is a homogeneously rough surface, the Monin–Obokhov (MO) similarity theory is a well-tested approach for specification of the boundary fluxes. This approach requires an empirically calibrated hydrodynamic roughness length to represent the aggregate effects of all flow-resisting surface details at unresolved scales, and a displacement height to represent a vertical shift in the velocity profiles. For cases in which the surface height distribution varies slowly enough in the horizontal direction that it is spatially resolvable in the LES, but where the height itself is small enough to fall below the first vertical grid point, special challenges arise. First, it is shown herein that prescription of a horizontally varying displacement height, in the context of MO similarity, is insufficient to produce realistic results in this unique application. A subsequent approach, which is relatively easy to implement, and can be used in LES without the need to use a "terrain-following" coordinate system or computationally expensive alternatives such as local grid refinement, is proposed. The approach is based on a form drag expressed in terms of the frontal area and the corresponding incoming momentum flux of the flow. The frontal area is evaluated in terms of the flow-normal gradient of the resolved height distribution. The proposed model, named the surface gradient-based drag (SGD) model, is applied to a variety of rough surfaces for which appropriate data exist. These surfaces are horizontally resolved, but vertically unresolved. The resulting mean velocity distributions are compared with available data and good agreement is observed. In addition, we apply the SGD model in flow over two synthetic multiscale surfaces with prescribed, power-law surface-height spectra. For these cases, first- and second-order flow statistics agree with established results. Future perspectives are outlined.

Keywords Large-eddy simulation · Resolved obstacles · Surface gradient

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1 Introduction

Understanding of the complex dynamic interactions between turbulent flows and obstacles is important for many applications. For example, atmospheric boundary-layer (ABL) flows are affected by residential and industrial structures, topography and vegetation; in these cases, the boundary condition fluxes and subsequent transport of momentum and scalars are crucially important. Other geophysical examples include ocean–atmosphere interactions and the ocean bottom boundary layer. Numerically reliable and physically realistic simulation methods are needed to address such flows.

For turbulent flow simulations near geometrically complex obstacles, there are several methods available. The immersed boundary method (IBM) is a robust approach in which objects are resolved on the simulation grid. In this method, an additional local forcing term is added to the momentum equation. At computational points immersed within the obstacle, the forcing term is specified such that the velocity is locally forced instantaneously to zero (Mittal and Iaccarino 2005; Iaccarino and Verzicco 2003). The IBM has been used in many scenarios, for example Kanda et al. (2004), Coceal et al. (2007) and Xie et al. (2008) performed high-resolution simulations of turbulent boundary-layer flows over an urban-like array of blocks. Chester and Meneveau (2007) used IBM to model atmospheric flows over fractal geometric shapes that resemble trees. Recently, Bhaganagar et al. (2004) and Bhaganagar (2008) used IBM within a direct numerical simulation (DNS) of flow over three-dimensional "egg carton" roughness elements. In their work, the height amplitude of the roughness elements was varied to study the subsequent hydrodynamic response. Others, such as Orlandi and Leonardi (2006) and Orlandi and Leonardi (2008), used IBM for the DNS of flow over two- and three-dimensional roughness elements.

For flow over slowly varying topography, the so-called "terrain-following coordinates" (Gal-Chen and Sommerville 1975a,b) method has proven successful. The method involves coordinate transformation such that the domain, including topography at the lower boundary, is transformed to a rectangular domain. The transformation results in significant changes to the governing transport equations. While the evaluation of additional terms imposes further computational complexity and cost, the method is nevertheless favourable due to the high accuracy that can be achieved, especially when the topography is smoothly varying. Example use of this method can be found in Sullivan et al. (2008).

In Nakayama et al. (2004), DNS over a surface built through the superposition of two sinusoidal waves was performed, where the amplitude and wavelength of the second mode was two fifths and one fifth of the first wave, respectively. The first and second modes were designed to represent resolvable and subgrid-scale surface details, respectively. Results from DNS were used to develop suitable models for frictional and pressure stresses, to be added to the filtered (large-eddy simulation, LES) momentum equations. Filtering of the flow domain affects the boundaries, and the frictional force term accounts for the roughness effects removed during filtering (i.e. the second mode). In similar work, Calhoun and Street (2001) performed extensive LES of channel flow in which the top and bottom boundaries were a flat and wavy surface, respectively.

In the ABL, vegetation is a common component of physical surfaces. With respect to numerical simulations, each computational cell may contain a larger cross-sectional frontal area exposed to flow than the frontal area of a computational cell, increasing hydrodynamic drag. The canopy-stress model was developed for treatment of this problem, with notable early contributions from Shaw and Schumann (1992) and Su et al. (1998). Later, the method was used by Brown et al. (2001) for ABL flows over surfaces without vegetation, when the lower boundary was an undulating roughened wavy surface. Relative to the IBM or

terrain-following coordinates, the canopy-stress model is much simpler to implement, with added versatility for flow over vegetation. The canopy-stress model is a parameterization that requires specification of a drag coefficient. It is known that the drag coefficient is related to obstacle shape, computational resolution, and Reynolds number, and experimental data are often used to specify this coefficient.

The aforementioned methods—IBM, terrain-following coordinates and canopy stress—allow spatial resolution of flow obstacles and topography. In applications of flow over homogeneously distributed roughness, individual flow-obstructing roughness elements are often not resolved. Instead, such elements are considered in a spatially averaged sense. In LES at high Reynolds number, where closure of the transport equations is achieved through the parameterization of the subgrid-scale (SGS) stress tensor, the effects of the surface are often parameterized through the Monin–Obokhov (MO) similarity theory (Monin and Obukhov 1954). Within the surface layer, where the turbulence follows the so-called law-of-the-wall, a logarithmic velocity profile exists (under neutral stratification) and it is through this profile that one relates the surface effects to the flow field. LES of flow over heterogeneous roughness has been studied for some time and a number of approaches have been developed. For instance, Avissar and Pielke (1989) developed parameterizations for the SGS influence of surface heterogeneity in mesoscale numerical simulations. Bou-Zeid et al. (2004, 2007) developed models for an effective roughness in large-eddy simulation of the ABL when the lower surface roughness varies locally in patches. They considered several variations of the patch distribution for testing of their generalized approach, which was incorporated with the MO similarity theory. These works place the hydrodynamic roughness length at the bottom of the domain (i.e. $x_3/H = 0$, where the index 3 denotes the vertical direction and H is a characteristic length scale of the problem). The MO similarity theory can be further generalized to account for regions in which the rough surface is vertically displaced, using a zero-plane displacement height. The displacement height simply represents the height at which net momentum extraction occurs (Jackson 1981; Kaimal and Finnigan 1994; Castro 2007). This approach was recently used, for example, in Chamecki et al. (2007, 2009), for the simulation of ABL flow over uniformly vertically displaced rough surfaces.

The above review has outlined the various options for modelling boundary-layer flows over surfaces with different flow-resisting obstacles or topography. In the present study, we focus on a special case that has not received attention in the literature: how to model flow over surfaces such as that shown in Fig. 1. In this sketch a surface $h(x, y)$ transect is shown, in addition to the horizontal (Δ_x) and vertical (Δ_z) discretization. The key aspect shown in Fig. 1 is that $h(x, y)$ is vertically not resolved since $h(x, y) < \Delta_z/2$. However, in the horizontal direction the surface is resolvable, at least at the level of a piecewise constant approximation shown using the dotted horizontal lines. For instance, for LES of the ABL using a domain that is 2 km high and with 64 vertical nodes, the cells are $\Delta_z \approx 32$ m high

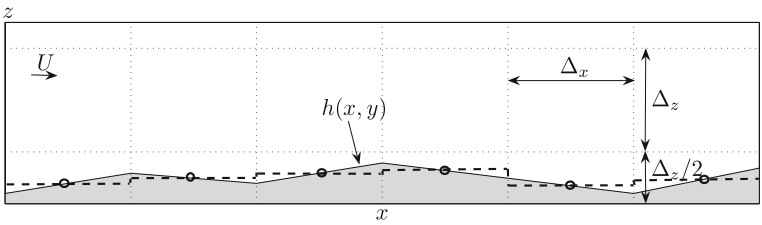

Fig. 1 Illustrative example of a horizontally resolved, but vertically unresolved, lower boundary

with the first node located at $z \approx 16$ m. If in the horizontal directions the domain is 4 km and horizontally resolved using 128 nodes, the horizontal grid width is $\Delta_x \approx 31.25$ m. We therefore envision surface features of height less than 16 m, and changing in the horizontal directions at scales on the order of 30–50 m.

We note here that such geometries may be represented in a LES code with local grid refinement (in all three directions), and/or using the standard IBM method. The latter method requires smooth velocity profiles near the wall, which is relevant to DNS or "wall resolving LES" (Pope 2000) at low Reynolds numbers when the viscous sublayer can be resolved. Our discussion in this paper instead refers to applications at high Reynolds numbers (non-wall resolving LES), in which also the use of local grid refinement is often not practical or desirable.

First, we will demonstrate that the use of a displacement height together with the MO boundary condition is not able, by itself, to account for this sort of horizontally resolvable roughness. This will be demonstrated using simulation results, and therefore Sect. 2 first describes the code and numerical details, and application of the displacement height to a representative test case. Section 3 describes the proposed model and implementation details. Section 4 presents the suite of test cases considered and the LES results, while the work is summarized in Sect. 5, with future perspectives outlined.

2 Large-Eddy Simulation Code

The LES code solves the incompressible high-Reynolds number momentum equation for pressure gradient driven boundary-layer flow (Albertson and Parlange 1999), and has been used extensively in the past (e.g., Bou-Zeid et al. 2005: Porté-Agel et al. 2000):

$$\frac{\partial \tilde{u}_i}{\partial x_i} = 0, \tag{1}$$

$$\frac{\partial \tilde{u}_i}{\partial t} + \tilde{u}_j \left(\frac{\partial \tilde{u}_i}{\partial x_j} - \frac{\partial \tilde{u}_j}{\partial x_i} \right) = -\frac{1}{\rho} \frac{\partial \tilde{p}}{\partial x_i} - \partial_j \tau_{ij} - \delta_{i1} \frac{1}{\rho} \frac{\partial P}{\partial x_1} + f_i, \tag{2}$$

where in Eq. 2, $\tilde{p}$ is dynamic pressure, f_i is a body force, $(\widetilde{\ldots})$ indicates filtering at the grid scale Δ, $\partial P/\partial x_1$ is the streamwise pressure forcing, and τ_{ij} is the SGS stress tensor. We consider neutrally stratified flow without the presence of a scalar. Incompressibility is enforced by the pressure Poisson equation, solution of which yields a pressure correction on the velocity field. The ABL is typically 1,000 m high, and under geostrophic flow conditions the Reynolds number is $O(10^8)$; in LES of such a flow over rough surfaces, viscous effects on momentum transport ($\nu \nabla^2 \tilde{\mathbf{u}}$) are safely omitted.

In LES, the deviatoric component of the SGS stress tensor is expressed using the eddy viscosity concept as:

$$\tau_{ij} - \frac{1}{3} \delta_{ij} \tau_{kk} = -2 \nu_t \tilde{S}_{ij}, \tag{3}$$

where ν_t is the turbulent eddy viscosity and $\tilde{S}_{ij}$ is the resolved strain-rate tensor. The well-known Smagorinsky model, $\nu_t = [C_S \Delta]^2 |\tilde{S}|$, where C_S is the Smagorinsky coefficient (Smagorinsky 1963), has seen several revisions. Early studies used location-, discretization-, and flow-dependent prescriptions for C_s (Lilly 1966; Deardorff 1970; Mason and Callen 1986; Piomelli et al. 1998). Germano et al. (1991) developed a dynamic tuning-free approach to computing the Smagorinsky coefficient, which was based on filtering the flow variables

at two resolutions: Δ and $a\Delta$, where $a > 1$ (generally 2), and assuming scale similarity between the information at these resolutions ($a\Delta$ is commonly referred to as the test-filter width). This approach has been further generalized and applied to other SGS parameterizations (Porté-Agel et al. 2000; Kosović 1997; Chow et al. 2005; Meneveau et al. 1996; Bou-Zeid et al. 2005). The SGS model used in this work is the Lagrangian scale-dependent dynamic model, presented by Bou-Zeid et al. (2005).

The height of the computational domain is H. The horizontal boundary conditions on the vertical faces of the domain are periodic, owing to the use of spectral methods in the horizontal directions. The horizontal domain dimensions, $L_x = L_y = L$, are varied between cases (x and y correspond to streamwise and transverse directions, respectively). The number of computational points in each direction are denoted as $N_x = N_y = N$, and N_z (z corresponds with the vertical). The domain is discretized such that $\Delta_x = L_x/N_x$, $\Delta_y = L_y/N_y$, and $\Delta_z = H/(N_z - 1)$. As the streamwise and transverse discretizations are equivalent, from this point we simplify by specifying $\Delta = \Delta_x = \Delta_y$. Discussion about specific cases is provided in Sect. 4.1.

The first vertical computational level is placed at height $\Delta_z/2$ from the ground ($z = 0$, Fig. 1). The code employs pseudo-spectral methods for evaluation of derivatives in the horizontal directions, while in the vertical direction derivatives are evaluated with centred second-order finite-differencing. Vertical velocities are placed on a staggered grid. The advection term of the momentum equation, $\tilde{u}_j \left(\partial\tilde{u}_i/\partial x_j - \partial\tilde{u}_j/\partial x_i\right)$ (Eq. 2), is de-aliased with the 3/2 rule (Orszag 1970). At the domain top, we impose the stress-free boundary condition by imposing $\partial\tilde{u}_i/\partial x_3|_{z=H} = 0, i = 1, 2$ where $i = 1$ and $i = 2$ denotes the streamwise and transverse directions, respectively. The flow is forced in the streamwise direction by an imposed pressure gradient,

$$\frac{\partial P}{\partial x_1} = -\frac{u_*^2\rho}{H}, \tag{4}$$

where u_* is the friction velocity and ρ is air density. The code is advanced in time with the second-order-accurate Adams–Bashforth scheme, and simulations are run with a non-dimensional timestep, $\Delta t^* = \Delta t u_*/H \approx 3.6 \times 10^{-4}$ (depending on case and time required for convergence). System variables are non-dimensionalized with domain height H and friction velocity u_*; the base hydrodynamic roughness is $z_0/H = 1 \times 10^{-6}$. All simulations are run until the flow field stabilizes with respect to various simulation statistics including the temporal evolution of kinetic energy.

2.1 Monin–Obokhov Similarity Theory and Displacement Height Method

In LES of the neutrally stratified ABL, MO similarity theory boundary conditions are used to model hydrodynamic drag on the flow. At the numerical grid points on the bottom surface, where wall-stress boundary conditions are specified, the streamwise and spanwise components of the wall-stress are specified according to Moeng (1984) and Mason (1994):

$$\tau_{i3}|_{\text{wall}}(x, y) = -\left[\frac{\kappa U^{\Delta}(x, y)}{\log\left(\frac{\Delta_z/2-d}{z_0}\right)}\right]^2 \frac{\widetilde{u}_i(\Delta_z/2)}{U^{\Delta}(x, y)}, \tag{5}$$

where $\widetilde{u}_i\,(\Delta_z/2) = \widetilde{u}_i(x, y, \Delta_z/2)$ is the local instantaneous velocity at the first grid-point above the surface ($z = \Delta_z/2$, for a staggered grid computational domain), and κ is the von Kármán constant ($\kappa = 0.4$). z_0 is the hydrodynamic roughness length, $d = h(x, y)$ is the displacement height set equal to the resolved height distribution, and

Table 1 Case domain details

Roughness element	Description	N	N_z	L/H	Comparison data
Block	Staggered, constant height	36	17	1.01	Coceal et al. (2007)
Block	Staggered, varying height	64	17	1.05	Xie et al. (2008)
Block	Uniform, constant height	36	17	1.01	Kanda et al. (2004): RUN 25
Ellipsoidal mounds	Uniform, constant height	64	33	1.10	Bhaganagar et al. (2004): $h^+ = 21.6$
Sinusoid	2 modes, 2-dimensional	60	33	1.26	Nakayama and Sakio (2002): case 2
RFM, $\beta_s = -0.8$	Multiscale surface	64	33	2π	NA
RFM, $\beta_s = -1.6$	Multiscale surface	64	33	2π	NA

$$U^{\Delta}(x, y) \equiv \left[\widetilde{u}_1 \left(x, y, \frac{\Delta_z}{2} \right)^2 + \widetilde{u}_2 \left(x, y, \frac{\Delta_z}{2} \right)^2 \right]^{1/2}, \tag{6}$$

is the local instantaneous resolved horizontal velocity magnitude at scale Δ. When computing the stress that will be substituted into the LES, the velocity field in Eq. 5 is test-filtered at scale 2Δ (i.e. $\widetilde{u}_i\,(\Delta_z/2) \rightarrow \widehat{\widetilde{u}}_i\,(\Delta_z/2)\,, i = 1, 2$, where $(\widehat{\widetilde{\ldots}})$ denotes test-filtering). Bou-Zeid et al. (2005) demonstrated that this test-filtering approach eliminates much of the undesired artifacts that arise from the fact that MO is valid for the mean flow, whereas it is used here locally without averaging (i.e. test-filtering removes only small-scale fluctuations but leaves larger-scale inhomogeneities). Note that $\tau_{13}|_{\text{wall}}$ and $\tau_{23}|_{\text{wall}}$ are in fact the $ij = 13$ and 23 components of the τ_{ij} tensor (Eq. 2, Sect. 2), but evaluated at $z = 0$.

Others (Chamecki et al. 2007, 2009) have successfully used the displacement height exclusively to represent a vertically displaced but horizontally homogeneous surface. Therefore, a first idea would be that perhaps specifying $\tau_{13}|_{\text{wall}}$ using Eq. 5 with a (x, y)-dependent displacement height (instead of a constant) might be appropriate to account for the additional drag experienced by the flow due to horizontally resolved, but vertically unresolved surface features. We have performed extensive LES tests with a multitude of different surfaces, with spatially non-homogeneous $d(x, y)$ in Eq. 5, all of which led to the same conclusion: specifying a non-homogeneous displacement height does not suffice to represent the effects of horizontal non-homogeneities of topography. To demonstrate this, we show results for the case of an urban-like staggered array of blocks, represented exclusively through the displacement height. The case is identical to that considered in Coceal et al. (2007), who compare their DNS results against existing experimental results for the same case (Cheng and Castro 2002), finding close agreement. Table 1 shows domain size and discretization details for this and other cases. The ratio of block height to height of the first computational level is $h_{\max}/(\Delta_z/2) = 0.9$. On the surface, a single computational point is allocated to each block, Fig. 2a illustrates the block array; that is to say $h(x, y) = h_{\max}$ if the grid point corresponds to the block, while $h(x, y) = 0$ if it is in between. For every four grid points, there is one point with $h = h_{\max} = 28.125 \times 10^{-3} H$ and three with $h = 0$. Comparisons with data are made in terms of the plane and time-averaged statistics (denoted by $\langle \ldots \rangle$), mainly the mean velocity profiles.

In Fig. 2b, we show both the results of Coceal et al. (2007) (circles) and LES results found when the block array is represented only through the displacement height (dashed). A classical logarithmic velocity profile for the specified roughness length is shown also (i.e. $\tilde{u}_1/u_* = 1/\kappa \log(z/z_0)$, dot-dashed). The profile demonstrates that the results of Coceal et al. (2007) are poorly reproduced by only adding an (x, y)-dependent displacement height

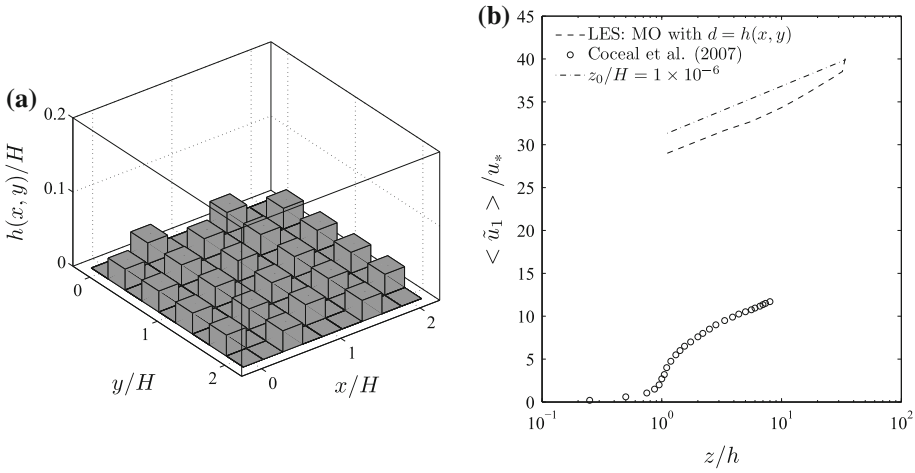

Fig. 2 Flow over the staggered block array case simulated by Coceal et al. (2007): **a** surface illustration; **b** velocity profile from literature dataset (*circles*) and LES (*dashed line*) in which the surface is represented only by the displacement height in the MO boundary condition. The *dot-dashed line* is the logarithmic law $\tilde{u}_1/u_* = 1/\kappa \log(z/z_0)$, valid for flows without the blocks

alone. Inclusion of the displacement height does impose additional drag, evidenced by a small vertical shift of the LES profile from the classical logarithmic law, but the added drag is far from sufficient. One notes that the profile from Coceal et al. (2007) in Fig. 2b displays a sharp change in shape in the near-surface region. This occurs because their DNS vertically resolves flow within the block canopy. We point out that above the blocks, the mean data from Coceal et al. (2007) could be represented by a roughness height of $z_{0_{[\text{Coceal}]}} \approx 8 \times 10^{-3} H$, i.e. much larger than the presently obtained value $z_{0_{\text{LES:MO}}} \approx 2 \times 10^{-6} H$ inferred from the LES results using only the displacement height (i.e. by fitting a line through the dashed line in Fig. 2b). The velocity profile is plotted against vertical height, where height is normalized by the local block height. In all subsequent velocity profiles shown in this work, we normalize height z by the maximum obstacle height, the only exception to this being when we compare with Xie et al. (2008), where we normalize with the average obstacle height.

3 Proposed Model for Horizontally Resolved Surfaces

The results in Fig. 2b motivated the development of a new model to account for spatial variations of the surface height. We propose a canopy stress-like model, in which the momentum of the incoming flow into a surface element frontal area is opposed by a force. The model is very simple to implement, requires limited computational expense and storage, and is effective.

The proposed model is added to the momentum equation using a body force per cell volume (f_i, Eq. 2), within the first computational mesh near the surface (Fig. 1). The model is a resolvable pressure drag, or form drag, model, accounting for obstacles and topography resolved at the LES grid-filter width, Δ, and described by $h(x, y)$. The model requires knowledge about the obstacle frontal area and cell volume. The domain is discretized in the vertical direction into cells of vertical height Δ_z, creating cells of volume $\Delta_x \Delta_y \Delta_z$. The momentum flux associated with the flow across a surface area A in general is given by $\rho \int_A \tilde{\mathbf{u}} \, (\tilde{\mathbf{u}} \cdot \mathbf{dA})$,

where $\tilde{\mathbf{u}}$ is the velocity vector and ρ is density. For the case shown in Fig. 1, the frontal elevation change along one cell length in the horizontal direction normal to the incoming flow is given by $\hat{\mathbf{n}} \cdot \nabla h \ \Delta_x$, where the velocity unit vector is used, given by $\hat{\mathbf{n}} = \tilde{\mathbf{u}}/U^{\Delta}$. Multiplying the elevation change with the width Δ_y gives the flow-frontal area, leading to a total momentum flux of $\rho \ \tilde{\mathbf{u}} \ U^{\Delta} \ \hat{\mathbf{n}} \cdot \nabla h \ \Delta_x \Delta_y$. For inclusion in the momentum equation we divide by density and local mesh volume, $\Delta_x \Delta_y \Delta_z$. The proposed drag model assumes that this momentum flux is exactly canceled by an applied body force near the surface. Furthermore, the opposing force (a form drag) should only act if the flow is 'into' the object, that is if $\hat{\mathbf{n}} \cdot \nabla h > 0$. Therefore, the model is expressed as:

$$f_i(x, y, \Delta_z/2) = -\frac{1}{\Delta_z} \widetilde{u}_i \ U^{\Delta} \ R\left(\hat{n}_k \frac{\partial h}{\partial x_k}\right), \tag{7}$$

for $i = 1, 2$, where the velocities are evaluated at a height $z = \Delta_z/2$, $\partial h/\partial x_k (k = 1, 2)$ are the horizontal components of the surface $h(x, y)$ first-order derivative, and $R(x)$ is the ramp function ($R(x) = (x + |x|)/2$) applied to the product of the surface topography derivative and velocity. We consider the incoming flow in both the streamwise and transverse directions, and specify that the obstacle only causes hydrodynamic drag when the flow component is wall normal into the obstacle; otherwise, we assume that there is no drag force since on downstream facing surfaces, often involving flow separation, the force is much smaller. This is achieved by application of the ramp function $R(x)$. An important aspect of the model is that, in all cases, the discretization is equivalent in both streamwise and transverse directions ($\Delta = \Delta_x = \Delta_y$); this constraint could easily be generalized for a non-uniform computation mesh. Equation 7 is not applied in the vertical direction (i.e. $f_3 = 0$, Eq. 2, to be consistent with implementation of the MO similarity theory, Eq. 5).

Combined with the displacement height through the standard MO boundary conditions, the model described here is designed to represent the sum of hydrodynamic drag effects in boundary-layer flow over a rough surface, $h(x, y)$. We refer to the proposed model as the surface gradient-based drag (SGD) model. To validate the approach, we again consider LES of flow over a staggered array of uniform height blocks, (see Coceal et al. 2007, Fig. 2a). In representing drag effects from partially resolved obstacles, friction drag effects are omitted. This is based on the very high Reynolds number characteristic of the ABL (Sect. 2).

4 Application of SGD Model to Various Flows

4.1 Cases

The literature contains a variety of datasets for flow over varying types of surface shapes and wall-mounted obstacles. We have selected the cases shown in Table 1, where the surfaces considered are all different. In Coceal et al. (2007), Xie et al. (2008) and Kanda et al. (2004), the cases are arrays of urban-like obstacles; case Xie et al. (2008) is a staggered array of spatially varying height blocks and Kanda et al. (2004) is a uniform array of constant height blocks. Visualization of surfaces are found in Figs. 2a, 6a, and 7a. From Kanda et al. (2004), we simulate the $\lambda = 0.25$ case, where λ denotes the areal coverage occupied by blocks as a fraction of horizontal domain area, $L_x L_y$ (in their work, they consider a variety of different λ). Recall that we specify the stress-free boundary condition at the domain ceiling, $\partial_3 \tilde{u}_{1,2} = \tilde{u}_3 = 0$ at $z = H$. Also, in our simulations the domain heights are chosen to be much higher than those of prior studies. This is because those studies aimed at resolving the near-surface features accurately, whereas the present study considers those features to be

"small", and below the first grid point. Therefore, when comparing with the data from Kanda et al. (2004), we use their channel-flow data up to the point at which their data display zero shear, i.e. $\partial \tilde{u}_1(z = H)/\partial x_3 \approx 0$.

The Bhaganagar et al. (2004) case is an "egg carton" surface of three-dimensional ellipsoidal roughness elements (see Bhaganagar 2008, and Fig. 8a for illustration of the surface). The Bhaganagar et al. (2004) result considered here is for Reynolds number $Re = u_* \delta/\nu = 400$, where δ is the channel half-height. The LES code used in the present work (Eq. 2, Sect. 2) neglects viscosity effects entirely, and therefore we might expect that the LES may over predict the streamwise velocity compared to the simulation of Bhaganagar et al. (2004) that includes viscous effects. The case of Nakayama and Sakio (2002) is a superposition of two sinusoidal modes (shown in Nakayama and Sakio 2002 and here in Fig. 9a). For all the cases, the domain height H and number of vertical grid points are chosen so that $\max(h(x, y)) = 0.9\Delta_z/2$.

In order to explore the applicability of the model to more random surfaces, we also use the new SGD model for LES over synthetic multiscale surfaces built using the random Fourier modes (RFM) method. The surface is constructed as,

$$h(\mathbf{x}) = \sum_{\mathbf{k}} c k^{\frac{1}{2}(\beta_s - 1)} e^{i(\mathbf{k}\cdot\mathbf{x} + \varphi)} \tag{8}$$

where c is an amplitude, β_s is the spectral slope, and φ is a random phase shift between 0 and 2π and chosen from a uniform distribution for each horizontal wavenumber $\mathbf{k}$. The surface is built at the LES resolution Δ, and so the sum extends over the appropriate wavenumbers. Here, we consider surfaces with spectral slopes $\beta_s = -0.8$ (more rough) and -1.6 (less rough). Domain details are shown in Table 1. Figure 11a and b illustrates the surfaces considered in this section, built with Eq. 8. We note that the difference between the highest and lowest values of $h(x, y)$ (a measure of amplitude) is equivalent between the surfaces. The imposed homogeneous value of z_0 for the MO boundary conditions and applied pressure gradient forcing (Eq. 4) is unchanged from the previous cases.

4.2 Results

Results are shown for the cases detailed in Table 1, in terms of the streamwise plane-averaged normalized velocity profiles. First, we apply the new model to the case of Coceal et al. (2007), a staggered array of blocks (Fig. 2a). Figure 3 shows the results of the model together with the results of Coceal et al. (2007), shown already in Fig. 2. Note that relative to the results in Fig. 2b, implementation of the new SGD model in conjunction with the displacement height imposes a much higher drag and yields much improved results. Note that the small uptick in mean streamwise velocity at the top of the domain is a numerical artifact of the top boundary condition under what is a very coarse vertical resolution and very small eddy viscosity there; this artifact is not expected to have any effects on the lower parts of the flow. Other results with the same vertical resolution show this uptick, while this effect disappears for LES with increased vertical resolution.

Second-order statistics for this case (with and without SGD model) are shown also in Figs. 4 and 5. Firstly, we draw attention to differences when using the SGD model, and when the blocks are represented by means of a displacement height only (case: MO with $d = h(x, y)$). Figure 4a illustrates that addition of the SGD model strongly suppresses the streamwise variance in the near-surface region, at the first grid point from the surface. Similar behaviour is observed with the transverse variance. In addition, we observe the relative influence of the SGD model in the vertical shear-stress profile (Fig. 5). When the SGD

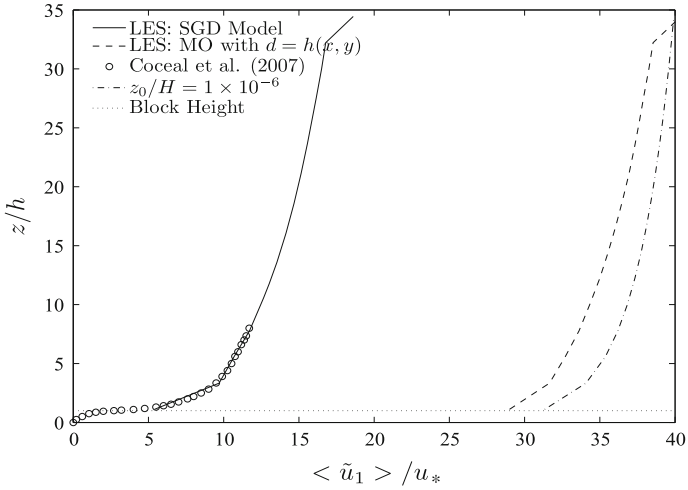

Fig. 3 Velocity profile from LES with and without SGD model and Coceal et al. (2007)

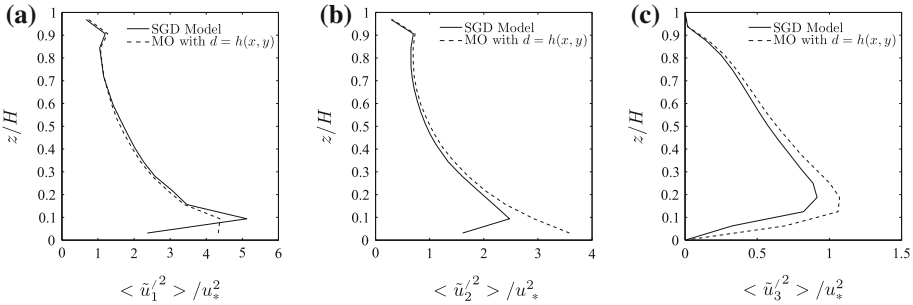

Fig. 4 Vertical distribution of normalized velocity variance: **a** steamwise $\langle \tilde{u}_1'^2 \rangle$; **b** transverse $\langle \tilde{u}_2'^2 \rangle$; **c** vertical $\langle \tilde{u}_3'^2 \rangle$

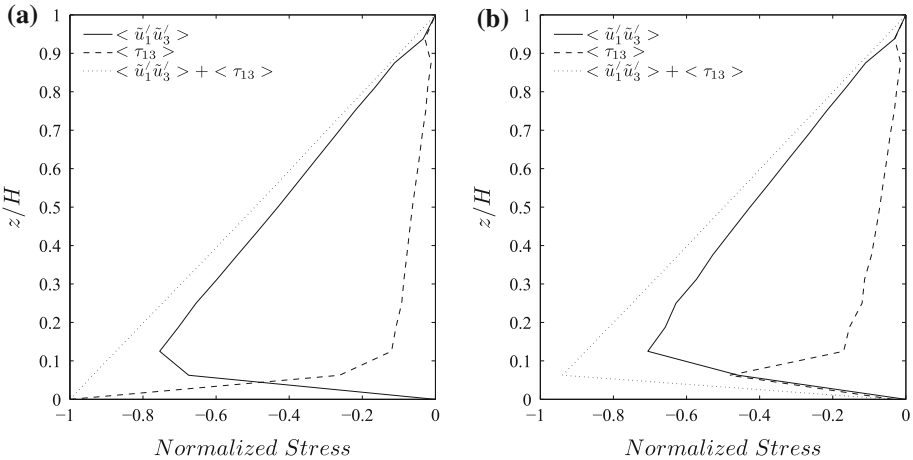

Fig. 5 Vertical distribution of normalized shear stress ($ij = 13$ components) for: **a** MO with displacement height; **b** SGD model

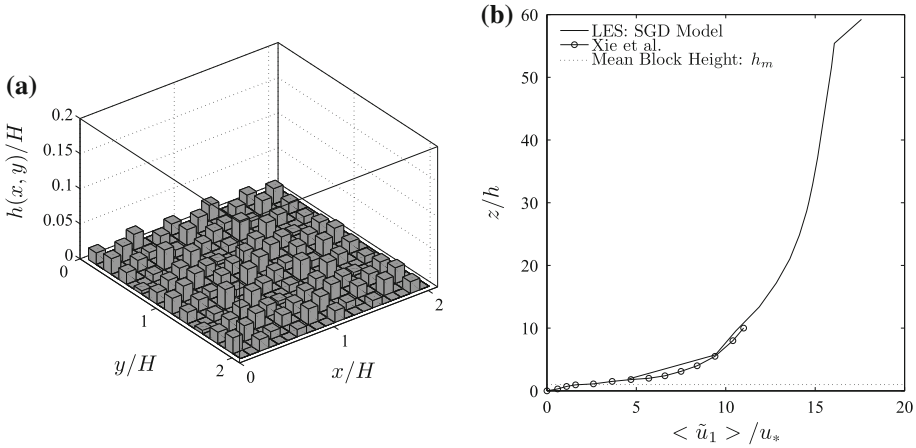

Fig. 6 Flow over a staggered array of varying-height blocks: **a** surface illustration; **b** velocity profile from Xie et al. (2008) (*circles*) and present LES using SGD model

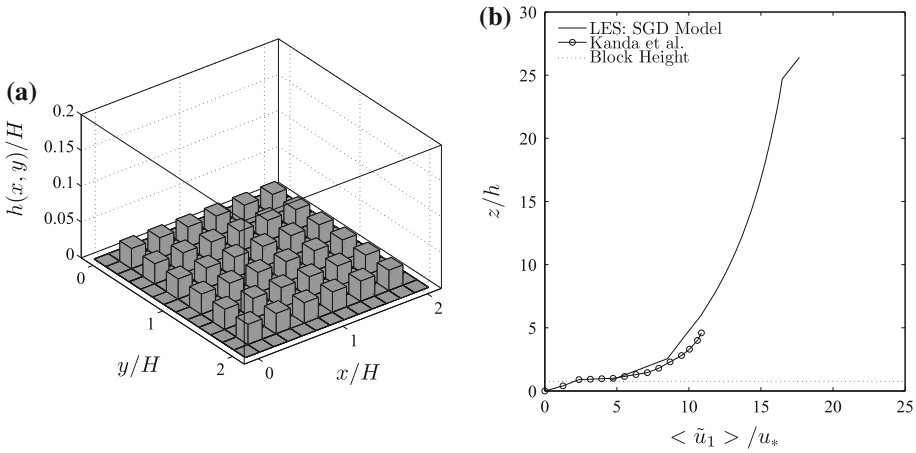

Fig. 7 Flow over uniform array of constant-height blocks simulated by Kanda et al. (2004) RUN 25: **a** surface illustration; **b** velocity profile from the literature dataset (*circles*) and present LES using SGD model

model is added in this case, the contribution of the MO wall-stress with displacement height (Eq. 5) is almost negligible. This indicates that the SGD model is imposing the majority of hydrodynamic drag (note that the contributions of the SGD term to the momentum flux are not indicated in this figure). Similarly, Leonardi and Castro (2010) reported that in DNS of channel flow over an array of cubes, drag effects were dominated by form drag.

When the SGD model is excluded, we observe the classical LES momentum flux profiles for flow over homogeneous roughness (Fig. 5a). Direct comparison with the variance data of Coceal et al. (2007) is difficult due to the different domain sizes. In general, due to the much smaller vertical domain size used in Coceal et al. (2007), their variances are lower than those obtained here. However, at $z/h = 1$, or $\Delta_z/2$ in our code, there is reasonable agreement (Fig. 7b from Coceal et al. 2007).

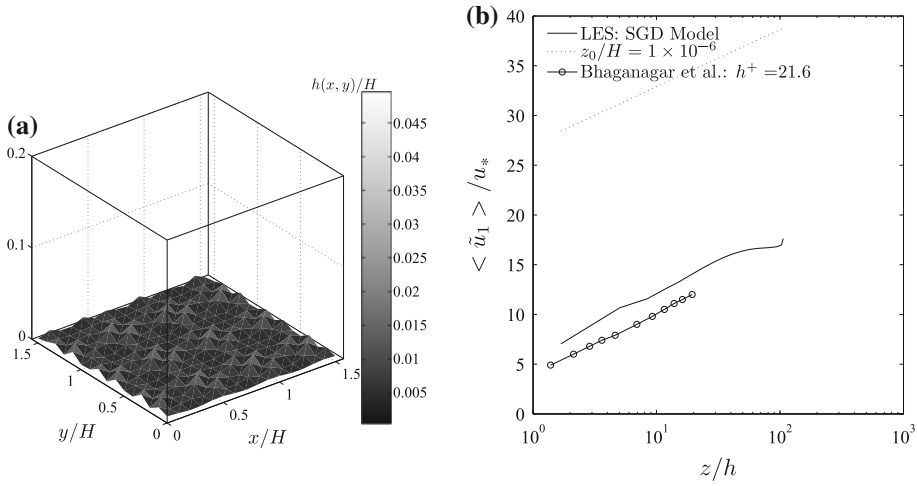

Fig. 8 Flow over regular array of three-dimensional ellipsoidal roughness elements simulated by Bhaganagar et al. (2004) $h^+ = 21.6$: **a** surface illustration; **b** velocity profile from constant height literature dataset (*circles*) and present LES using SGD model

Figures 6b through 9b show velocity profiles for the remaining cases listed in Table 1, with data from the literature shown also. The comparison data are taken directly from the mean velocity plots normalized by wall friction velocity as given in the literature, with the exception of the case of the uniform array of constant height blocks, Kanda et al. (2004). For this case, the friction velocity was not given and we obtained it from their reported plane-averaged momentum flux profile (Kanda et al. 2004: Fig. 7). In all cases, the agreement between the data and the new SGD model is quite reasonable. The LES velocity agrees closely with the data for flow over the staggered array of blocks of spatially varying height (Xie et al. 2008) and flow over a regular array of constant height blocks (Kanda et al. 2004), shown in Figs. 6b and 7b, respectively. Figure 8b illustrates that for flow over the so-called "egg-carton" roughness, the LES velocity slightly exceeds that in the dataset of Bhaganagar et al. (2004). In Sect. 4.1 we already mentioned our expectation that the present LES velocity would exceed the Bhaganagar et al. (2004) data, as their simulations are for much lower Reynolds number and thus include additional frictional drag. Finally, our SGD LES and literature data agree closely for flow over the two-sinusoid surface Nakayama and Sakio (2002) case, Fig. 9b. Recall that we assume all drag is imposed from the momentum flux into a frontal area, with no further adjustments. Obviously the results could be further improved by additional adjustments, but the goal is to propose and test a simple and generally applicable, approximate, and practical method.

Figure 10 shows a profile of the transverse average of the streamwise velocity at the first vertical computational level ($\Delta_z/2$), across the two-sinusoid (Nakayama and Sakio 2002, Fig. 9a) surface case; a transect of the surface (not-to-scale in the vertical) is superimposed on the figure for clarity (streamwise flow is from left-to-right). The velocity field used for this figure was averaged during the last 30% of the simulation. The profile shows the near-wall velocity accelerating in the "lee" of the surface peak (i.e. $0 \leq x/H \approx 1$), and subsequently decelerating as the boundary layer approaches the next peak of the surface ($1 \leq x/H \leq 2$). Figure 10 illustrates a clear interaction of the flow field with the surface, demonstrating the influence of the SGD model.

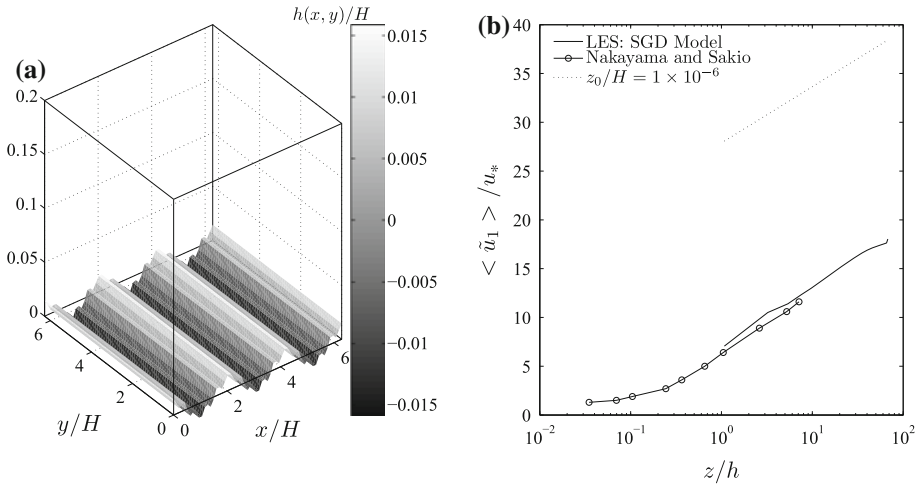

Fig. 9 Flow over two-sinusoid wavy surface simulated by Nakayama and Sakio (2002), case 2: **a** surface illustration; **b** velocity profile from the literature dataset (*circles*) and present LES using SGD model

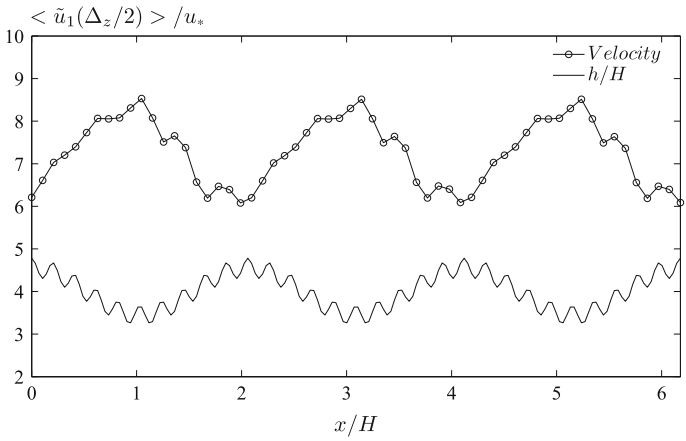

Fig. 10 Near-wall velocity $\langle \tilde{u}_1(\Delta_z/2) \rangle / u_*$ from LES of the Nakayama and Sakio (2002), case 2 dataset. Surface transect shown not-to-scale in the vertical

4.3 Results for Random Fourier Modes Surface

The streamwise mean velocity for the RFM surfaces described in Sect. 4.1, and illustrated in Fig. 11, is shown in Fig. 12. For both surfaces, the expected logarithmic law behaviour is observed in the near-surface region; the velocity profile indicates correct addition of drag evidenced by a vertical shift of the velocity profile below the original logarithmic law profile (dot-dashed line). We also observe that the mean velocity magnitude for the $\beta_s = -0.8$ surface is smaller than the velocity for $\beta_s = -1.6$, as expected since the $\beta_s = -0.8$ surface is more rough. The effective roughness height $z_{0,\text{eff}}$ measured for both cases is $z_{0,\text{eff}} = 0.225 \times 10^{-3} H$ for $\beta_s = -0.8$ and $z_{0,\text{eff}} = 0.095 \times 10^{-3} H$ for $\beta_s = -1.6$.

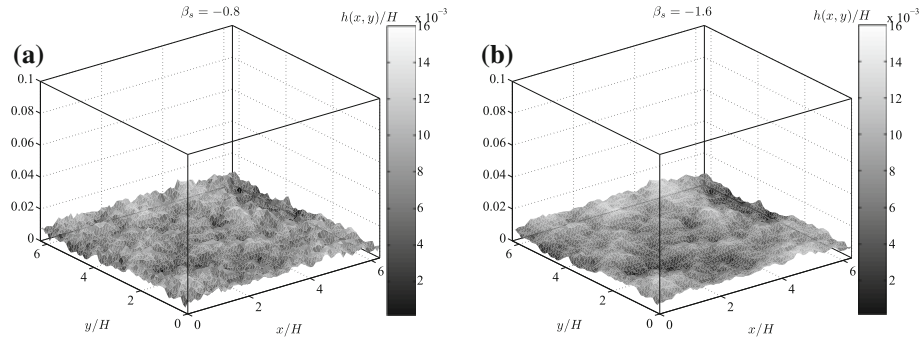

Fig. 11 Surface height field $h(x, y)$ for: **a** $\beta_s = -0.8$; and **b** $\beta_s = -1.6$

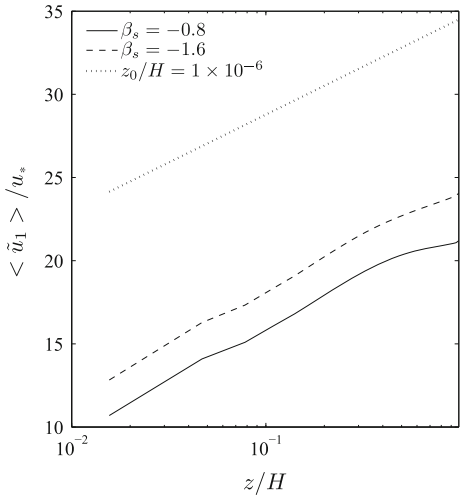

Fig. 12 Non-dimensional mean streamwise velocity profiles on a semi-logarithmic scale for RFM surfaces

The root-mean-square (rms, σ) values of the surfaces are $2.07 \times 10^{-3} H$ and $2.3 \times 10^{-3} H$, respectively, so we find that the effective roughness length for the entire surface is $z_{0,\text{eff}} = \alpha\sigma$, with $\alpha = 0.11$ and 0.04 for $\beta_s = -0.8$ and -1.6, respectively. The strong dependence of α on the type of surface represents a challenge to formulate effective roughness models and has motivated us to develop a dynamic tuning-free approach to predict such roughness, particularly for surfaces with roughness modes larger and smaller (subgrid-scale) than the horizontal resolution, Δ_x.

Second-order statistics are shown in Figs. 13 and 14. Figure 13 shows plane-averaged vertical distributions of the components of simulated velocity variance. These values and profiles are comparable to those reported elsewhere, for example in Porté-Agel et al. (2000) and Anderson et al. (2007). The vertical distribution of normalized stress component $ij = 13$ is shown in Fig. 14. For both surfaces, we see that the first data point for total stress is not equivalent to the non-dimensional pressure gradient (Eq. 4), -1, but correctly reduces monotonically towards the domain top after the peak value. The total normalized stress is decomposed into resolved and SGS stress tensor contributions. The contribution from the SGS model $\langle \tau_{13} \rangle$ is highest in the near-surface region just above the surface, and reduces monotonically towards the domain top.

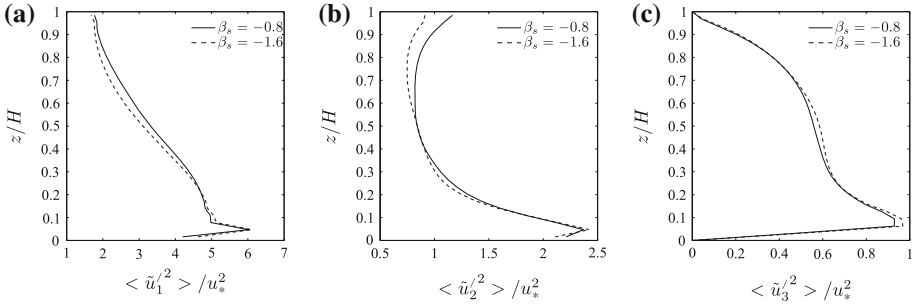

Fig. 13 Vertical distribution of normalized velocity variance: **a** steamwise $\langle \tilde{u}_1'^2 \rangle$; **b** transverse $\langle \tilde{u}_2'^2 \rangle$; **c** vertical $\langle \tilde{u}_3'^2 \rangle$

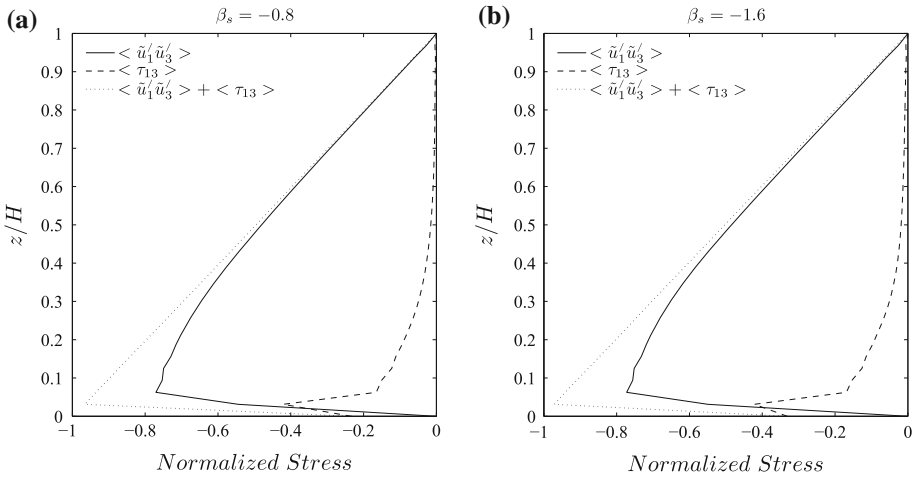

Fig. 14 Vertical distribution of normalized shear stress ($ij = 13$ components) for: **a** $\beta_s = -0.8$; **b** $\beta_s = -1.6$. Within each figure, the components of stress are decomposed according to the legend

The streamwise velocity profile (Fig. 12) indicates that the $\beta_s = -0.8$ surface imposes a higher hydrodynamic drag than the $\beta_s = -1.6$ surface. This is expected, since a flatter spectral slope is indicative of a rougher surface. This implies that the SGD model effect (discussed in Sect. 4.4) is more pronounced for $\beta_s = -0.8$. In the absence of the SGD model, the total wall-stress should be equivalent to the system forcing (i.e. $\langle \tau_{13}|_{\text{wall}} \rangle = -1$ at $z/H = 0$). The Reynolds stress vanishes at the surface. However, Fig. 14 shows that $\langle \tau_{13}|_{\text{wall}} \rangle > -1$, illustrating that the SGD model imposes the balance of the remaining wall stress. In the following section, we confirm this and show that the SGD model contributes the correct balance of remaining wall-stress. Other users of canopy-stress models (e.g., in Brown et al. 2001: Fig. 4) observe similar behaviour.

4.4 Wall-Stress Components

For illustration only, the boundary conditions may be collectively described through combination of the standard logarithmic formulation (Eq. 5) and integration of the volumetric force associated with the SGD model, such that:

Table 2 Wall-stress budget

Comparison case	$-\langle\tau_{13}\vert_{\text{wall}}\rangle$	$-\Delta_z\langle f_1\rangle$	$-\langle\tau_{13}^T\rangle$
Coceal et al. (2007): ($f_i = 0$)	1.0	0	1.0
Coceal et al. (2007): ($f_i \neq 0$)	0.007	0.98	0.99
Xie et al. (2008)	0.010	1.002	1.012
Kanda et al. (2004): RUN 25	0.007	0.98	0.99
Bhaganagar et al. (2004): $h^+ = 21.6$	0.074	0.906	0.98
Nakayama and Sakio (2002): case 2	0.072	0.92	0.992
RFM, $\beta_s = -0.8$	0.233	0.793	1.026
RFM, $\beta_s = -1.6$	0.334	0.689	1.023

$$\tau_{i3}^T(x, y) = \tau_{i3}|_{\text{wall}}(x, y) + \int_0^{\Delta_z} f_i(x, y, z)dz = \tau_{i3}|_{\text{wall}}(x, y) + \Delta_z f_i(x, y, \Delta_z/2), \quad (9)$$

where $i = 1, 2$, $f_i(x, y, \Delta_z/2)$ is given by Eq. 7, and superscript T denotes total.

For all the cases considered in our work, Table 2 shows a budget of the wall-stress boundary conditions (components of Eq. 9). In all cases, we see that the total stress is equivalent to the non-dimensional pressure gradient (system forcing, -1). Interestingly, we observe that for the RFM surfaces, the contribution from the SGD model $\Delta_z\langle f_1\rangle$ increases with flatter spectral slope. This is consistent with Figs. 12 and 14, showing a higher drag for flatter spectral slope. Additionally, it can be argued that the surfaces with ellipsoidal (Bhaganagar et al. 2004) and sinusoidal (Nakayama et al. 2004) roughness elements are geometrically more similar to the RFM surfaces than the urban-like block array cases (Kanda et al. 2004; Coceal et al. 2007; Xie et al. 2008). Table 2 accordingly shows that the SGD model contributions for the Bhaganagar et al. (2004) and Nakayama et al. (2004) cases are comparable to the RFM, while the SGD model contribution is much larger than the MO contribution for the block configurations. This is an expected result, as the urban-like array of blocks (Kanda et al. 2004; Coceal et al. 2007, and Xie et al. 2008) imposes significantly higher hydrodynamic drag than undulating surfaces. The plots of normalized shear stress for this case (Fig. 5) show clearly the effect of adding the SGD model, which completely dominates over the MO effects (the same is shown here in Table 2).

Substitution of the MO and SGD model equations into Eq. 9 leads to,

$$\tau_{i3}^T(x, y) = -\left[\left(\frac{\kappa}{\log\left(\frac{\Delta_z/2-h}{z_0}\right)}\right)^2 + R\left(\hat{n}_k\frac{\partial h}{\partial x_k}\right)\right]\tilde{u}_i U^{\Delta}(x, y). \quad (10)$$

This rewriting of the force as a stress boundary condition is only applicable if the force model is applied at the boundary of the computational domain. It shows that the proposed SGD model may be equivalently expressed as a local roughness length equal to,

$$z_{\text{tot}}(x, y) = \left(\frac{\Delta_z}{2} - h\right)\exp\left[-\kappa\left(R\left(\hat{n}_k\frac{\partial h}{\partial x_k}\right) + \left(\frac{\kappa}{\log\left(\frac{\Delta_z/2-d}{z_0}\right)}\right)^2\right)^{-\frac{1}{2}}\right]. \quad (11)$$

5 Conclusions

Numerical techniques such as the IBM or "terrain-following coordinates" are useful in boundary-layer flow over surfaces where flow-resisting roughness elements are spatially well resolved within the computational mesh in all three directions. However, if the lower boundary is composed of low-amplitude topography or wall-mounted obstacles of limited height, such that they can be resolved in the horizontal directions but not in the vertical, the aforementioned methods are not useful or may be too complicated to be implemented in practice. There are numerous situations where this would occur, for example, in ABL flow over "rolling", fluvial topography. One simple approach to representing the effects of such a surface in LES would be with a spatially varying displacement height, within the MO similarity theory. However, our results show that modulating the displacement height has only a small influence on the flow, as illustrated by representing an urban-like array of staggered blocks within an LES, and comparing with existing observations in the literature.

In our study, a pressure drag SGD model for use in LES has been proposed. This simple and user-friendly model is applicable when the surface is horizontally resolved, but vertically unresolved. The formulation and implementation of the model is somewhat similar to the canopy-stress model. Unlike the canopy-stress model, here the gradient of surface topography is used to isolate the component of the surface thought to be hydrodynamically relevant (i.e. frontal area normal to the local flow). The model does not rely on a specified drag coefficient. Instead, we assume a force equal and opposite to the momentum flux in an area equivalent to the frontal area faced by incoming flow. We note that this assumption is tantamount to retaining the classical pre-factor for form drag, $\frac{1}{2}C_d$, and setting $C_d = 2.0$; we note a case from the literature where $C_d = 2.0$ was reported at $z/h = 1.0$ for flow over an urban-like staggered array of blocks (Coceal et al. 2006, Figure 20i).

The SGD model is subsequently tested with a set of entirely different surfaces. In comparing the results, plane-averaged streamwise velocity profiles are used. The agreement is not always exact, but better than 10% except for the low Reynolds number egg carton surface case, in which viscous effects could be expected to affect the results. Related efforts centre on combining the proposed SGD model with a dynamic model for subgrid-scale height fluctuations, for cases where the surfaces also display broad unresolved horizontal small-scale fluctuations.

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